

IIS & DNS - (CCNA Networking)

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8:31 AM

Server:

A machine which provides a collection of services such as data storage, DHCP, etc., is known as a Server.

There are different types of Servers based on structure.

1. Rack Server
2. Break Server

Server also contains NIC and it can be assigned with IP's.

Each Server consists of an IP which is public and static.

Collection of Servers is known as a Data Centre.

Types of Servers:

1. Web Server (IIS) - It hosts the websites
2. File Server
3. Database Server
4. Application Server
5. Mail Server
6. DNS Server

Types of Services:

1. Web Hosting Service - allows clients to host and access websites via the web.
2. File Sharing Service - enables the sharing of files between different users or systems. (e.g., using Samba or NFS)
3. Database Service - provides database access, query processing, and management for applications (e.g., SQL services).
4. Mail Service - enables sending, receiving, and managing emails (e.g., using SMTP, IMAP, POP# protocols).
5. Authentication Service - provides user verification (e.g., Active Directory, LDAP).
6. Remote Access Service - Allows users to access a network or serve remotely (e.g., VPN services, RDP).

When you install a service on a server, that server takes on the role of providing that service. For example, if the DNS service is installed on a server, that server becomes a DNS server.

OS:

=> Operating System

There are two types of OS based on which a device is recognized as a Client or a Server. They are,

1. Client - it includes,
 - 1) Win xp
 - 2) Win 2007
 - 3) Win 8 & 8.1
 - 4) Win 10
 - 5) Win 11

2. Server - it includes,
 - 1) Win 2008
 - 2) Win 12 & 12r
 - 3) Win 2016
 - 4) Win 2020
 - 5) Win 2020

Devices with client OS are client and devices with server OS are servers

PC's and Laptops are designed with client OS in default. PC's and Laptops can also have server OS but they cannot handle the requests and responses since they have low specifications.

IIS:

Internet Information Services is a flexible, secure, and manageable web server developed by Microsoft.

It is used to host websites and web applications, enabling communication between a web server and a client (usually a web browser).

IIS runs on the Windows Server operating system.

Port no: 80

Problem--

Since there n number of web services and websites, to reach them we cant remember all of the IP's.

Solution--

Name resolution --> DNS

For example,

Think about the contacts saved in your mobile phone. Remembering hundreds of phone numbers is difficult, so we save them with names such as "John" or "Alice."

When we want to call someone, we simply select their name from our contacts instead of typing their 10-digit phone number. The phone then looks up the corresponding number and connects the call.

DNS works in a similar way on the Internet. Instead of remembering IP addresses (such as 142.250.183.78), we use easy-to-remember website names (such as google.com). DNS translates the website name into its corresponding IP address, allowing your device to connect to the correct server.

In this analogy:

- Contact Name → Domain/Website Name
- Phone Number → IP Address
- Mobile Contact List → DNS Database
- Finding the phone number from the contact name → DNS Name Resolution

- IP of Google.com - 172.217.24.14
- Address of DNS - 10.254.171.198